Using lab-01 image to reproduce the analysis in the article: ‘Joint modeling of social genetic effects in mono- and pluri-specific groups: case study in intercrops’

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2026-03-25

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Overview

A prebuilt Docker image (**lab-01**) containing R, all required packages, and system dependencies needed to reproduce the analyses in this repository is available on DockerHub.

1. Prepare a project folder

Create a folder for the nalysis (e.g. `paper1-simul`) and copy the repository files/folder (as organized) into it.

2. Verify Docker

Make sure Docker is installed on your system (https://docs.docker.com/desktop/):

```
docker --version
docker info
```

3. Pull the prebuilt image

Pull the image from DockerHub (this image is public — normally, no Docker Hub account is required to pull it):

```
docker pull jemaysalomon/lab-01:0.1.12
```

4. Run an interactive container

Mount your project folder when launching the container so you can work on the files. Note that this mirrors the project files — any changes made inside the container will be reflected in the local files. Make sure you mount to the correct pre-created folder.

```
docker run --rm -it \
  -v ~/path/to/paper1-simul:/home/project \
  jemaysalomon/lab-01:0.1.12 bash
```

You can check the R version and list installed packages. See the Docker CLI reference for more. You can also install additional packages and customize the image to suit your needs.

```
R --version
Rscript -e "installed.packages('your-package')"
```

5. Run the analysis (e.g. the plots)

Edit the path in `Plots.Rmd`. You should be one directory level up in R, so the path becomes:

```
projectDir <- here::here("../")
```

Then run the following in the terminal:

```
Rscript -e "rmarkdown::render('R/Plots.Rmd', output_dir='R')"
```

You can repeat the same steps for the other files.

Note that, upon acceptance of this article, this Docker image will contain code, data, and supplementary files, everything needed to fully reproduce the analyses in a single, self-contained environment.

Contact & License

For any questions regarding this tutorial, send an email to: jemaysalomon@gmail.com.

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